

Workshop 4.1: Postprocessing

Release 2020R2



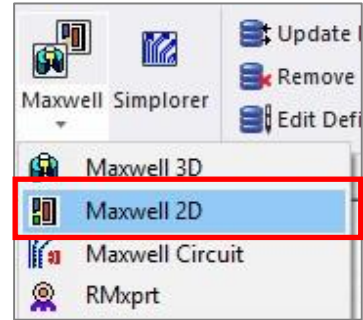
Overview

- **Post Processing in Maxwell 2D**
 - **This workshop will discuss how to use the Maxwell 2D Post Processor. Field plots and calculator operations will demonstrated on an Eddy Current project. The following tasks will be performed:**
 - Plot the mesh
 - Plot of MagB
 - Plot of B_vector
 - Plot of MagH along a line
 - Create a table of MagH along a line
 - Calculate average MagB in an object
 - Verify $\text{DivB} = 0$
 - Calculate loss in the conductor
 - Calculate net and total current flowing in the conductor
 - Export field results to a file.

Model Setup

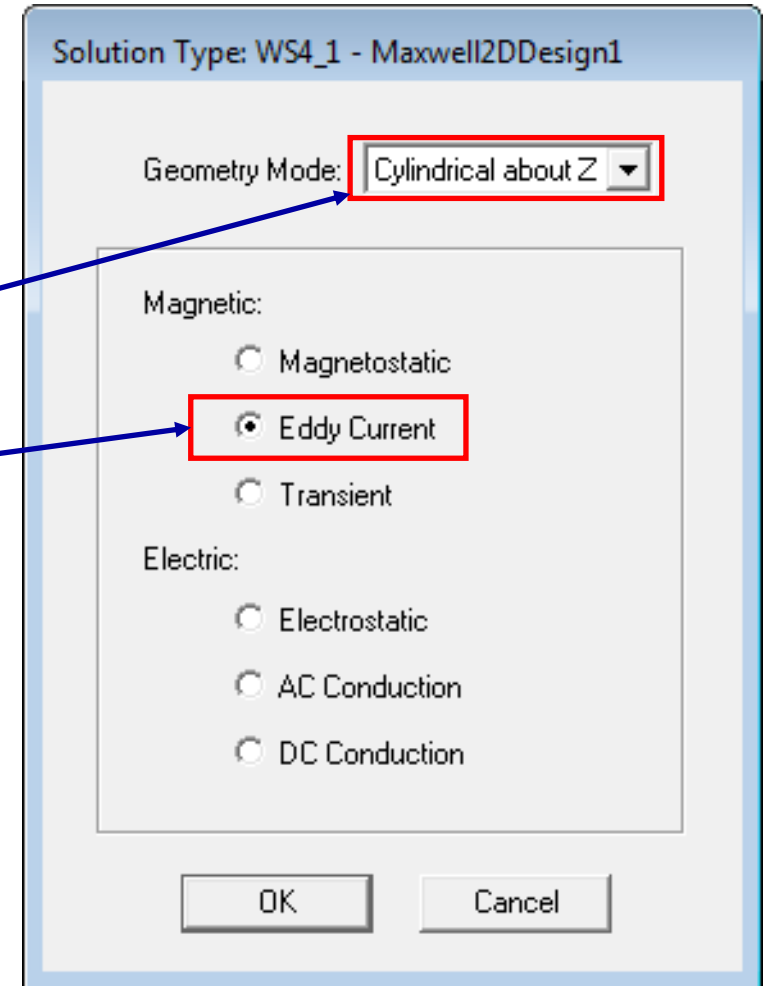
- Create Design

- Select the menu item **Project** → **Insert Maxwell 2D** Design, or click on the icon in drop down list Maxwell on panel Desktop
- Name the Design **WS_4_1**



- Set Solution Type

- Select the menu item **Maxwell 2D** → **Solution Type**
- Solution Type Window:
 - Geometry Mode: **Cylindrical about Z**
 - Choose **Magnetic > Eddy Current**
 - Click the OK button

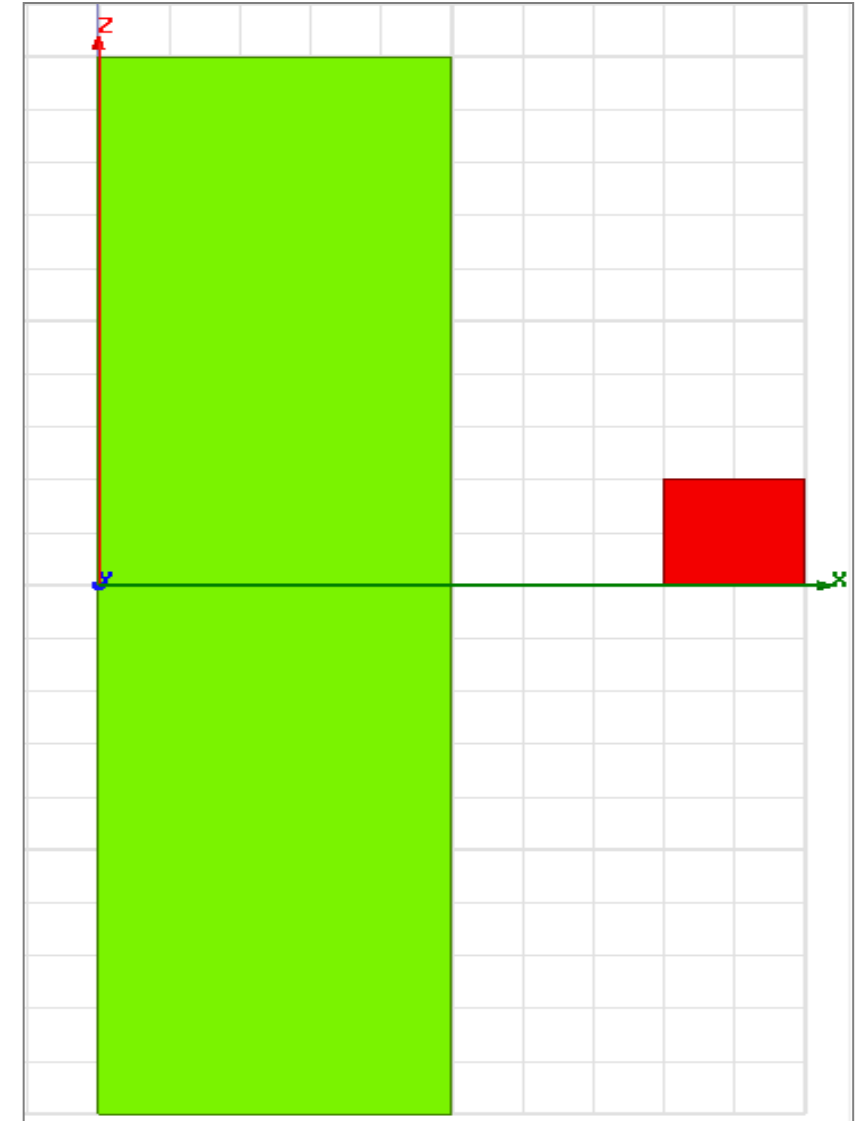


- Set Model Units

- Select the menu item **Modeler** → **Units**
- In Set Modeler Units window,
 - Select units: **mm** (millimeters)
 - Press the OK button

Create Model

- Create Core
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the center of the base
X: 0, Y: 0, Z:-10, Press the **Enter** key
 - Using the coordinate entry fields, enter the radius
dX: 5, dY: 0, dZ: 20, Press the **Enter** key
 - Change the name of the sheet to **Core** and material to **iron**
- Create Coil
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the center of the base
X: 8, Y: 0, Z: 0, Press the **Enter** key
 - Using the coordinate entry fields, enter the radius
dX: 2, dY: 0, dZ: 2, Press the **Enter** key
 - Change the name of the sheet to **Coil** and material to **copper**

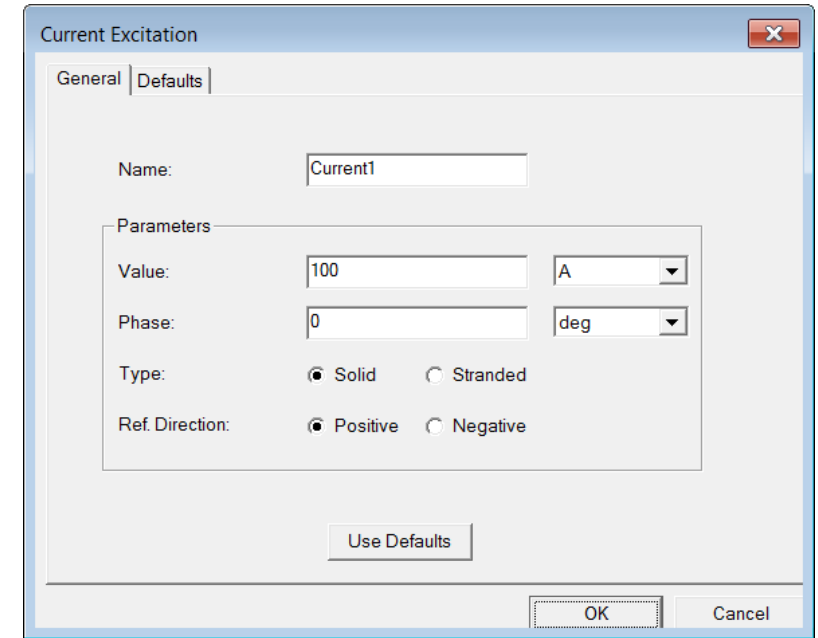


Create Model

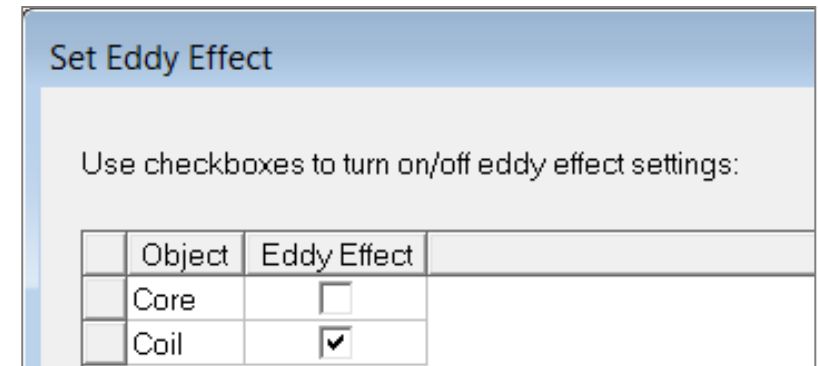
- Create a line for plotting of the fields
 - Select the menu item *Draw → Line*
 - Using the coordinate entry fields, enter the first vertex
X: 0, Y: 0, Z: 0, Press the *Enter* key
 - Using the coordinate entry fields, enter the second vertex
X: 0, Y: 0, Z: 20, Press the *Enter* key
 - Press Enter to exit line creation
- Create Simulation Region
 - Select the menu item *Draw → Region*
 - In Region window,
 - Padding all directions similarly: ☒ *Checked*
 - Padding Type: Percentage Offset
 - Value: *100*
 - Press OK

/ Assign Excitation

- Assign Excitation
 - Select the sheet Coil from the history tree
 - Select the menu item **Maxwell 2D → Excitations → Assign → Current**
 - In Current Excitation window,
 - Value: **100 A**
 - Phase: **0 deg**
 - Type: **Solid**
 - Press OK
- Set Eddy Calculations
 - Select the menu item **Maxwell 2D → Excitations → Set Eddy Effect**
 - In Set Eddy Effect window,
 - **Core**
 - Eddy Effects: ☐ **Unchecked**
 - Press OK



The 'Current Excitation' dialog box is shown with the 'General' tab selected. The 'Name' field contains 'Current1'. The 'Parameters' section includes: 'Value' set to 100 with units 'A', 'Phase' set to 0 with units 'deg', 'Type' set to 'Solid' (radio button selected), and 'Ref. Direction' set to 'Positive' (radio button selected). A 'Use Defaults' button is located below the parameters. At the bottom right are 'OK' and 'Cancel' buttons.



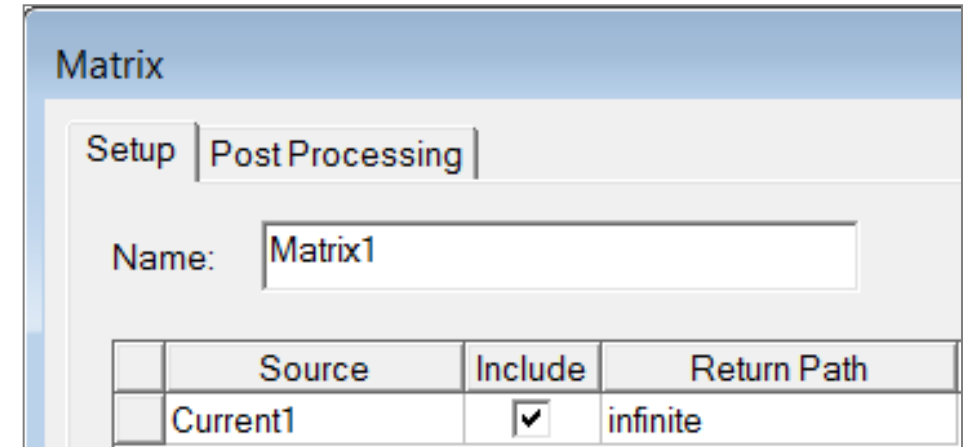
The 'Set Eddy Effect' dialog box is shown. It contains the instruction 'Use checkboxes to turn on/off eddy effect settings:'. Below this is a table with two columns: 'Object' and 'Eddy Effect'.

Object	Eddy Effect
Core	☐
Coil	☒

Note: Eddy Effects can be calculated only with solid excitations. For Stranded coils eddy effects are not considered.

Assign Parameters and Boundary

- Setup Impedance Calculations
- Select the menu item *Maxwell 2D → Parameters → Assign → Matrix*
- In Matrix window,
 - **Current1**
 - Include: ☒ Checked
 - Press OK
- Assign Boundary
 - Select the sheet **Region** from history tree
 - Select the menu item *Edit → Select → All Object Edges*
 - Select the menu item *Maxwell 2D → Boundaries → Assign → Balloon*
 - In Balloon Boundary window,
 - Press OK



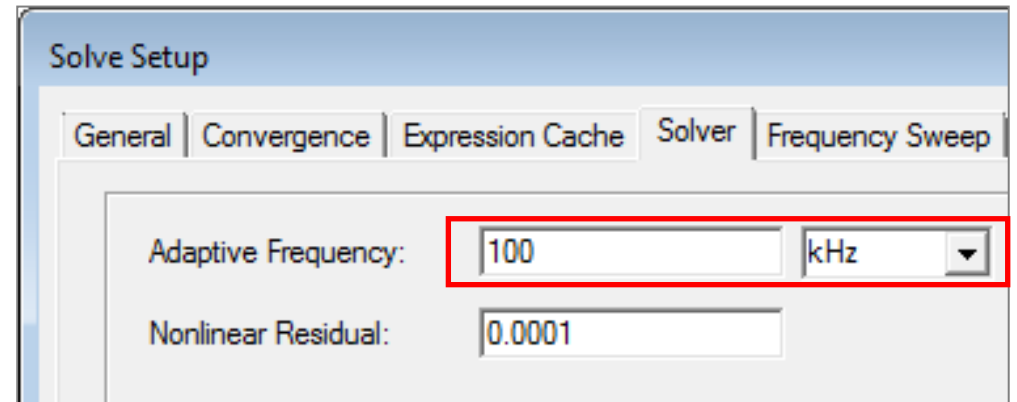
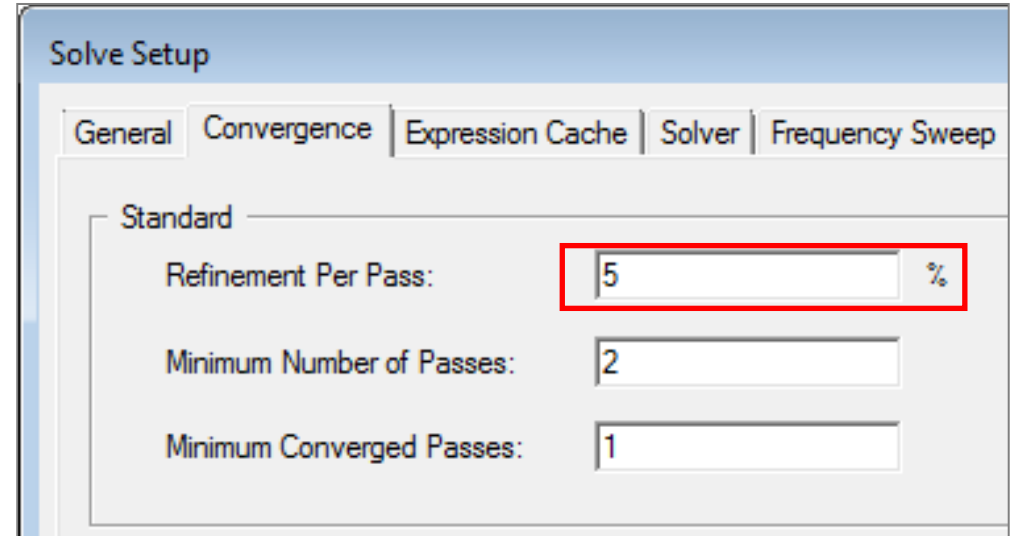
Note: On symmetry axis, “Balloon Boundary” assignment is automatically skipped. This can also be achieved by selecting the edges of region which are not on symmetry axis.

Apply Mesh Operations

- Apply Mesh Operations for Core
 - Select the sheet Core from the history tree
 - Menu item *Maxwell 2D → Mesh Operations → Assign → Inside Selection → Length Based*
 - In Element Length Based Refinement window,
 - Name: *Length1*
 - Restrict Length Of Elements: ☐ Unchecked
 - Restrict the Number of Elements: ☒ Checked
 - Maximum Number of Elements: 500
 - Press OK
- Apply Mesh Operations for Coil
 - Similarly assign mesh operation to Coil with following parameters
 - Name: *Length2*
 - Restrict Length Of Elements: ☐ Unchecked
 - Restrict the Number of Elements: ☒ Checked
 - Maximum Number of Elements: 500
 - Press OK

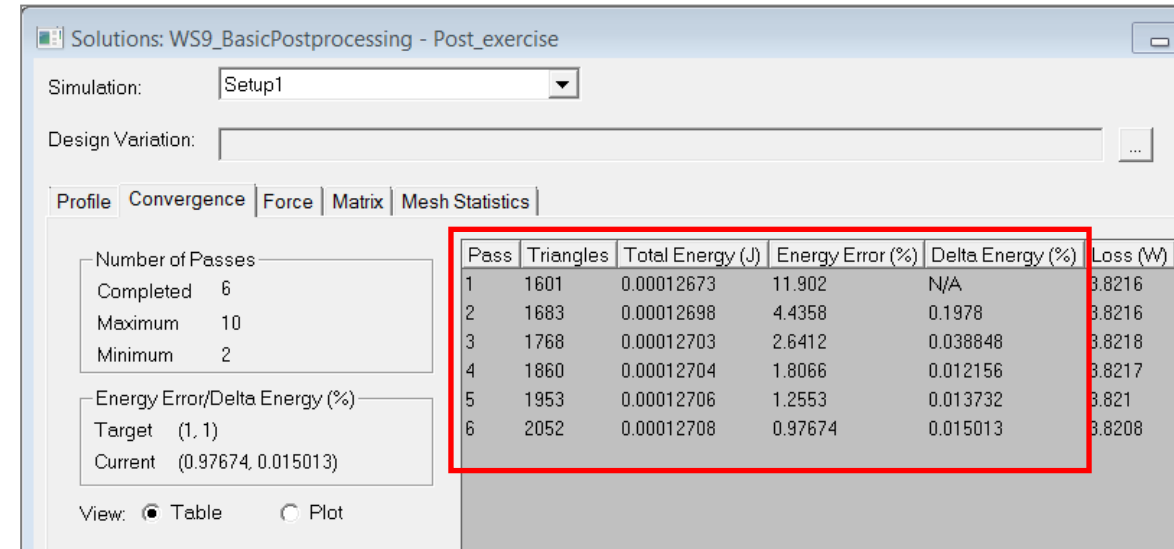
Analyze

- Create an analysis setup:
 - Menu item **Maxwell 2D** → **Analysis Setup** → **Add Solution Setup**
 - Solution Setup Window:
 - **Convergence tab**
 - Refinement Per Pass: 5 %
 - **Solver tab**
 - Adaptive Frequency: 100 kHz
 - Click the OK button
- Run Solution
 - In Project manager window **RMB on Setup1** → **Analyze**



Solution Data

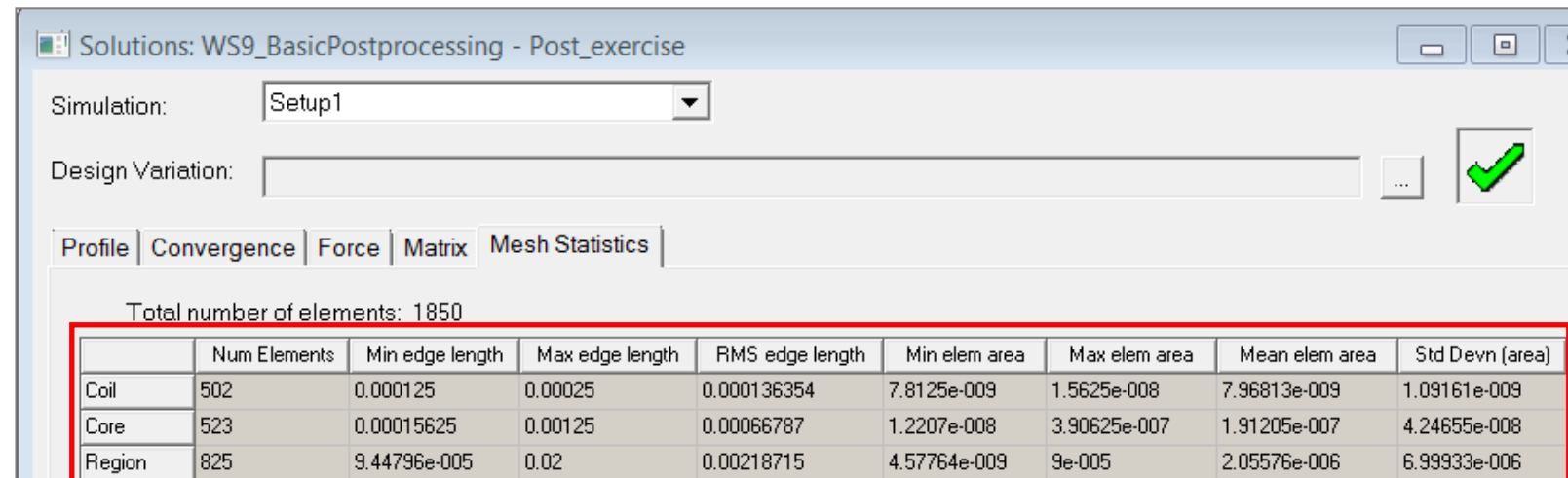
- View the Solution Data:
 - Menu item **Maxwell 2D → Results → Solution Data**
 - To view the Profile:
 - Click the **Profile Tab**
 - To view the Convergence:
 - Click the **Convergence Tab**



Pass	Triangles	Total Energy (J)	Energy Error (%)	Delta Energy (%)	Loss (W)
1	1601	0.00012673	11.902	N/A	3.8216
2	1683	0.00012698	4.4358	0.1978	3.8216
3	1768	0.00012703	2.6412	0.038848	3.8218
4	1860	0.00012704	1.8066	0.012156	3.8217
5	1953	0.00012706	1.2553	0.013732	3.821
6	2052	0.00012708	0.97674	0.015013	3.8208

Note: The default view for convergence is Table. Select the Plot radio button to view a graphical representations of the convergence data.

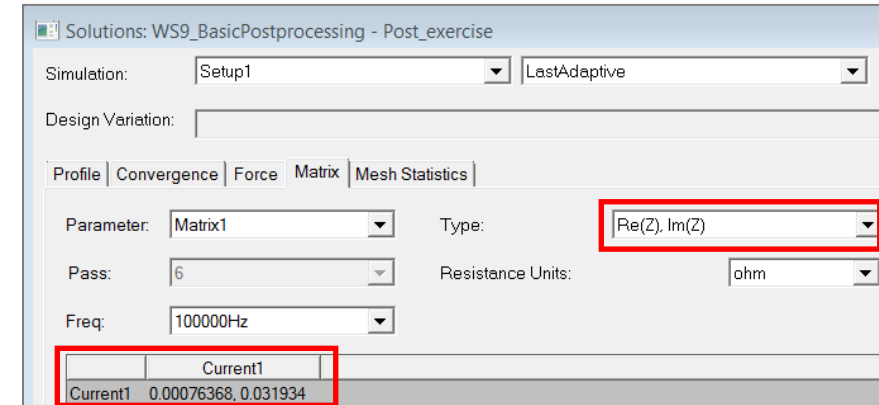
- To View Mesh information
 - Click **Mesh Statistics Tab**



	Num Elements	Min edge length	Max edge length	RMS edge length	Min elem area	Max elem area	Mean elem area	Std Devn (area)
Coil	502	0.000125	0.00025	0.000136354	7.8125e-009	1.5625e-008	7.96813e-009	1.09161e-009
Core	523	0.00015625	0.00125	0.00066787	1.2207e-008	3.90625e-007	1.91205e-007	4.24655e-008
Region	825	9.44796e-005	0.02	0.00218715	4.57764e-009	9e-005	2.05576e-006	6.99933e-006

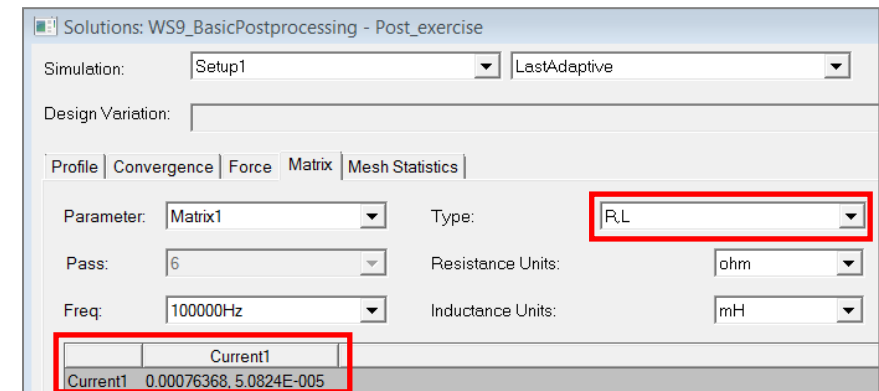
Solution Data

- View Matrix Results
 - To View Impedance values
 - Click Matrix tab
 - Set Type to Re(Z), Im(Z)



Note: the imaginary term of the matrix includes only inductive reactance (Ohms)

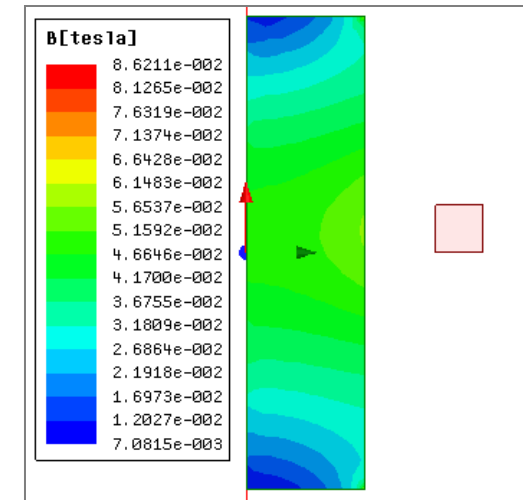
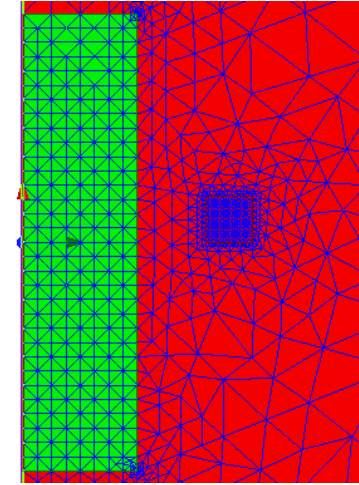
- Change Type to R,L



Note: the imaginary term of the matrix includes only the inductance (Henry)

Create Field Plots

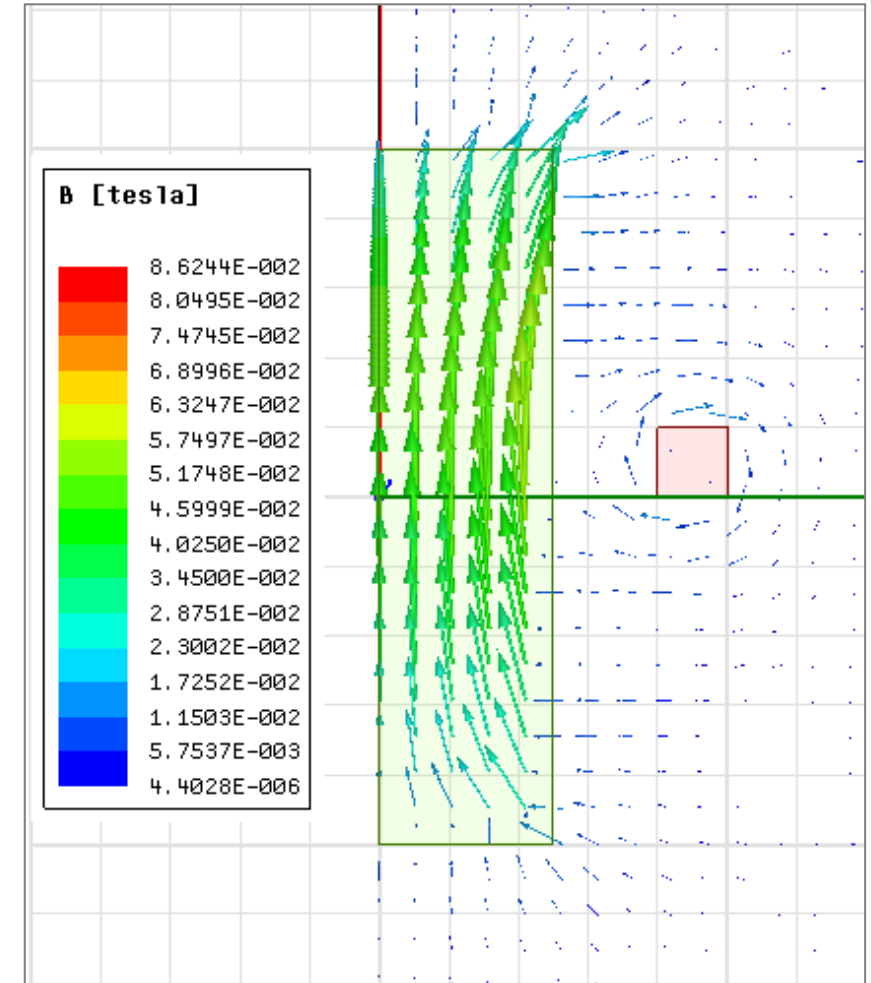
- Plot Mesh
 - Select the menu item **Edit** → **Select All**
 - Select the menu item **Maxwell 2D** → **Fields** → **Plot Mesh**
 - In Create Mesh Plot window,
 - Press Done
- Plot Mag_B
 - Select the sheet Core from the history tree
 - Select the menu item **Maxwell 2D** → **Fields** → **Fields** → **B** → **Mag_B**
 - In Create Field Plot window,
 - Press Done



Note: To hide the previously created field plots, right click on plot from Project manager window and uncheck “Plot Visibility”

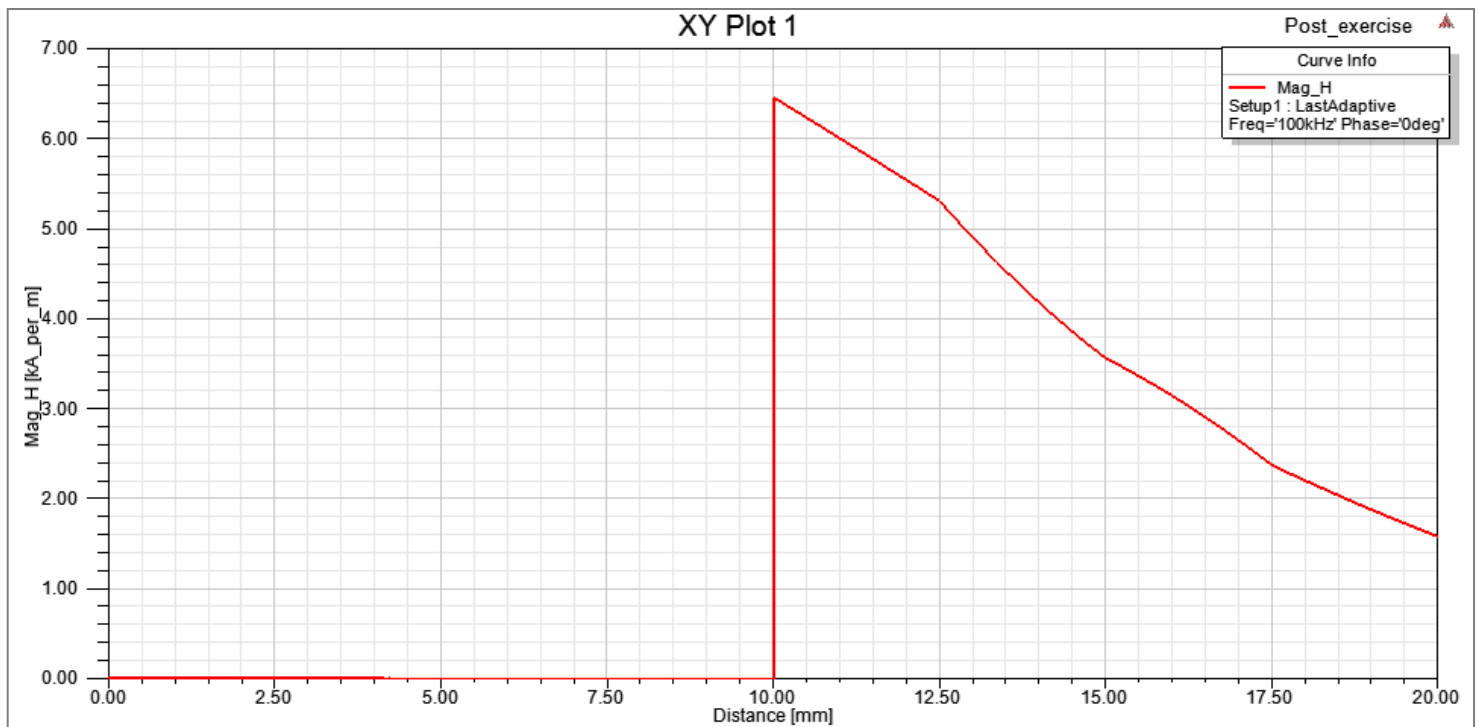
Create Vector Plot

- Plot Vector_B
 - Select the menu item *Edit* → *Select All*
 - Select the menu item *Maxwell 2D* → *Fields* → *Fields* → *B* → *B_Vector*
 - In Create Field Plot window
 - Press Done
 - Double click on the legend to change plot properties
 - In the window
 - *Plots* tab
 - Plot: B_Vector1
 - Select the tab OnSurface
 - Vector Plot
 - Spacing: Five Spaces
 - Min: 0.5
 - Max: 2
 - Press Apply and Close



Create XY Plot

- Plot Mag_H along a Line
 - Select the menu item **Maxwell 2D** → **Results** → **Create Field Reports** → **Rectangular Plot**
 - In **Report** window.
 - Geometry: Polyline1
 - X : Default
 - Category: Calculator Expressions
 - Quantity: Mag_H
 - Select New Report



Create Data Table

- Table of Mag_H along a Line
 - Select the menu item **Maxwell 2D → Results → Create Field Reports → Data Table**
 - In **Report** window
 - Geometry: Polyline1
 - X: Default
 - Category: Calculator Expressions
 - Quantity: Mag_H
 - Select New Report

	Distance [mm]	Mag_H [kA_per_m] Setup1 : LastAdaptive Freq='100kHz' Phase='0deg'
1	0.000000	0.009644
2	0.020000	0.009645
3	0.040000	0.009647
4	0.060000	0.009649
5	0.080000	0.009651
6	0.100000	0.009653
7	0.120000	0.009654
8	0.140000	0.009656
9	0.160000	0.009658
10	0.180000	0.009659
11	0.200000	0.009661
12	0.220000	0.009663
13	0.240000	0.009664
14	0.260000	0.009666
15	0.280000	0.009667
16	0.300000	0.009669
17	0.320000	0.009670
18	0.340000	0.009671
19	0.360000	0.009673
20	0.380000	0.009674
21	0.400000	0.009675

/ Calculate Average Mag_B

- Calculate Average Mag_B in an Object
 - Select the menu item *Maxwell 2D* → *Fields* → *Calculator*
 - Select Input > Quantity > B
 - Select General > Complex > Real
 - Select Vector > Mag
 - Select Input > Geometry
 - Select Volume > Core > Press OK
 - Select Scalar > Integral > RZ
 - Select Input > Number
 - Select Scalar > Value: 1 > Press OK
 - Select Input > Geometry
 - Select Volume > Core > Press OK
 - Select Scalar > Integral > RZ
 - Select General > / (Divide)
 - Select Output > Eval
 - The average value of flux density calculated is around **0.038 Tesla**

Scl : 0.0382180167956709

Scl : /(RZIntegrate(Volume(Core), Mag(Real(<Br,0,Bz>))), RZIntegrate(Volume(Core), 1))

/ Calculate DivB

- Calculate the Divergence of B-field
 - In Field Calculator window,
 - Select the button Clear to remove previous expression
 - Select Input > Quantity > B
 - Select General > Complex > Real
 - Select Vector > Divg
 - Select Input > Geometry
 - Select Volume > Core > Press OK
 - Select Scalar > Integral > RZ
 - Select Output > Eval
 - The divergence of B is approximately zero (numerical noises)

```
Scl: 1.31606806992419E-011  
Scl: RZIntegrate[Volume(Core), Divg(Real(<Br,0,Bz>))]
```

/ Calculate loss in a conductor

- Calculate losses in Coil
 - Select the button Clear to remove previous expression
 - Select Input > Quantity > OhmicLoss
 - Select Input > Geometry
 - Select Volume > Coil > Press OK
 - Select Scalar > Integral > RZ
 - Select Output > Eval
- The ohmic loss in the coil at 100kHz is approximately **3.82 Watt**

```
Sc1: 3.82072139800436  
Sc1: RZIntegrate(Volume(Coil), Ohmic-Loss)
```

Calculate current in a conductor

- Calculate Net Current in Coil
 - Select the button Clear to remove previous expression
 - Select Input > Quantity > J
 - Select General > Complex > Real
 - Select Vector > Scal? > ScalarPhi
 - Select Input > Geometry
 - Select Surface > Coil > Press OK
 - Select Scalar > Integral > XY
 - Select Output > Eval
- The net current in the coil is equal to the source amp-turns = +/- 100

Note: Since total current needs to be calculated over the 2D section of Coil, Integration is done with XY integral and not RZ

```
ScI: 100.0000000168649  
ScI: Integrate(Surface(Coil), ScalarPhi(Real(<0,Jphi,0>)))
```

/ Calculate current in a conductor

- Calculate Total Current including Eddy Component
 - Select the button Clear to remove previous expression
 - Select Input > Quantity > J
 - Select General > Complex > CmplxMag
 - Select Vector > Scal? > ScalarPhi
 - Select Input > Geometry
 - Select Surface > Coil > Press OK
 - Select Scalar > Integral > XY
 - Select Output > Eval
- Calculated value of current comes out to be around **130 A**

ScI : 129.92220449958

ScI : Integrate(Surface(Coil), ScalarPhi(CmplxMag(<0,Jphi,0>)))

Export Field Results

- Export Results to a file
 - Select the button **Clear** to remove previous expression
 - Select Input > Quantity > H
 - Select Output > Export
 - In Export Solution window,
 - Output file name: H_Vector
 - Calculate grid points: ☒ Checked
 - Minimum: (0, 0)
 - Maximum: (5, 5)
 - Spacing: (1, 1)

Export Solution

Output file name:

Grid points on which to export ☒ Include points in output file

☐ Input grid points from file

☒ Calculate grid points

Coordinate System: ☒ Cartesian ☐ Cylindrical ☐ Spherical

Offset mm mm mm

	Minimum	Maximum	Spacing
X	0 mm	5 mm	1 mm
Y			
Z	0 mm	5 mm	1 mm

Note: In order to use the field results in a different software, you can export the fields on a uniform 2-dimensional grid.

Saving the Project

- This completes the workshop
- Save the file with the name **Workshop_4_1** in the working folder



End of Presentation